

# Essential Tools for Reproducible Research

Pipelines, Containers & Workflow Management Systems

---

# | The Challenge in Bioinformatics Research

## Computational Chains

Distinct, often interdependent computational steps → Failure in one script affects the entire pipeline.

## Environmental Drift

Software versions, operating systems, and installed libraries can be inconsistent → **significant variance in results.**

# | So... What Can We Do?



## Pipelines

Defining the automated sequence of your research steps.



## Containers

“Freezing” the software environment to ensure consistency.



## Workflow Management Systems

Execute workflow across computational resources.

# | Pipelines: What & Why?

A specific set of programs or scripts run in a **predefined, automated sequence**.

- **Reduce Manual Errors:** Eliminates human mistakes
- **Saves Time:** Re-run complex analyses instantly on new datasets.
- **Detailed Records:** Step-by-step history of your data's history.

# Example: Metagenomics Analysis

```
@SRR26872690.1 M04209:105:000000000-KBJ
13:5423 length=148
CGTCCACCATACCAGCCGCGGAAGCATCATCCCCGCCC
ACGCTCAGGAAATGCAGCAGCAAGATAATCACGAGTATC
GCGGCAGACTTGCCACCAAGTCCAACCAATCAAGCAAC
+SRR26872690.1 M04209:105:000000000-KBJ
13:5423 length=148
```

Metagenomic  
Reads



MetaPhlAn



Bowtie2



DIAMOND



Taxonomic  
Profiling



Pangenome  
Search



Translated  
Alignment



Pathway  
Reconstruction



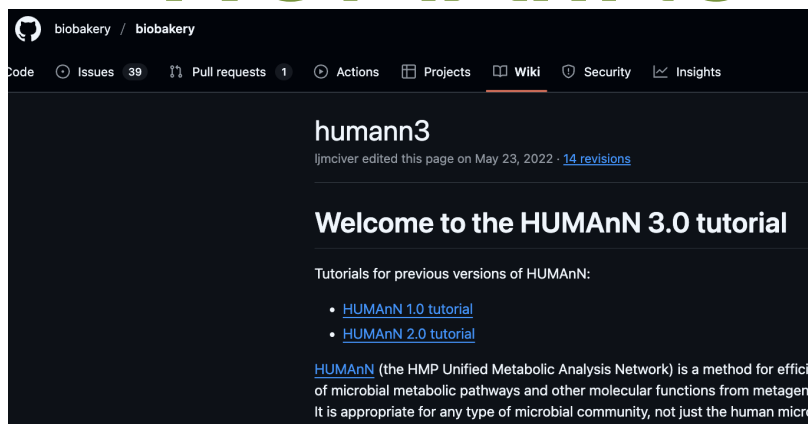
Output



MinPath



## HUMAnN3



# | Containers

## Keeping things constant!

- Self-contained, executable packages that bundle code, libraries, and settings.
- Containers **solve the "dependency hell" problem** by ensuring your analysis runs identically, anywhere, anytime.



# | Key Container Tools



**Docker**

The most widely used platform for developing and running apps in isolated environments. Ideal for local and cloud work.



**Apptainer**

Formerly Singularity. Designed for HPC and shared clusters where security and root-less execution are required.

# | Example Docker Container

## Elements:

- **Base Operating System:** Ubuntu (Version 24.04 LTS)
- **R Version** → R version 4.3.2
- **R Packages** → ggplot2 version 3.4.4, ggplot version 1.1.4
- **R Script** → my\_analysis.R
- **Filesystem** → the layered structure combining all elements above

**How do we keep our pipelines and  
containers organized?**

# | Workflow Management Systems

**Elements** A software framework that helps you build and execute pipelines (often using containers)

- Manages dependencies and tasks
- Handles resource distribution
- Scales computation across CPUs/Nodes
- Improves project organization

 nextflow → Scaling Science

**Highly scalable and portable pipelines for clusters and cloud platforms.**



Resources ▼

About ▼

## EPI2ME Workflows

September 01, 2020 · 3 min read



Where can we **store & share** our own pipelines,  
containers, and workflows?

# | Introducing GitHub

## A Web-Based Platform

Built around **Git**, GitHub is an online infrastructure for scientific collaboration and storage



A remote location for tracked files



Visual interface for project history



Secure cloud-based backup

# | GitHub: The Central Hub



## Version Control

A cloud-based platform to **store files with full history tracking.**



## Collaboration

Enables teams to review code, track issues, and manage scientific tasks in a **single centralized environment.**



## Provenance

Ensures every result can be **traced back to the exact version** of the code used during processing.

# Beyond Version Control



## A Research Suite

-  Documentation: READMEs and Wikis
-  Project Management
-  GitHub Pages: Host lab websites, tutorials

## Setting up Git

[https://ak-inbre-dsc.github.io/dsc\\_workshop\\_2026/tutorials/Git\\_Startup/](https://ak-inbre-dsc.github.io/dsc_workshop_2026/tutorials/Git_Startup/)

# | GitHub as the Glue: workflows

## Seamless Integration

GitHub integrates directly with Workflow Management System tools like Nextflow. You can run a pipeline directly from a repository with a single command:

```
nextflow run user/pipeline-repo
```

*“Well, it worked on my machine!”*

— Famous Last Words in Research

# | Lesson 2 & 3: Using and Collaborating with Github

## DSC\_Module\_3\_Lesson2\_Git\_Github.ipynb

- Set up a Github Account
- Connect your local Git repo to your Github

## DSC\_Module\_3\_Lesson3\_Collaborating...ipynb

- **Pair up** and collaborate on the same Github repo
- Notebooks are **labeled by collaborator identity** (1 or 2)

## Bonus: DSC\_Module\_3\_Lesson4\_RStudioServer\_Git.ipynb

- Using Git/Github with Rstudio Server
- Link & instructions in Resources doc